Advanced AWS Auto Scaling Group Setup Guide

Step 1: Create a VPC (Virtual Private Cloud)

• Go to the VPC Console → Create VPC. • Name tag auto-generation: Enter a name (e.g., project-demo). • Number of Availability Zones: 3. • Public Subnets: 3. • Private Subnets: 0. • NAT Gateway: None. • VPC Endpoints: None. • Click Create VPC to finalize.

Step 2: Configure Security Groups

• Go to the Security Groups console. • Choose your VPC ID. • Configure inbound and outbound rules: - Inbound: Allow HTTP (80), HTTPS (443), and SSH (22) as required. - Outbound: Allow all traffic (default). • Save your security group.

Step 3: Launch an EC2 Instance

• Go to EC2 → Launch Instance. • Name: e.g., siva-instance. • Amazon Machine Image (AMI): Amazon Linux 2023 (Kernel 6.12 AMI). • Instance type: t3.micro. • Key pair: Select an existing key (e.g., linuxxx). • Network: Select your VPC (project-demo). • Auto-assign Public IP: Enable. • Attach Security Group: Select the default or custom group created earlier. • Advanced Details → User Data: Paste your startup script if needed. • Launch the instance, and once running, test it using its public IP in your browser.

Step 4: Create an Amazon Machine Image (AMI)

• Go to EC2 → Instances. • Select your running instance. • Actions → Image → Create Image. • Image Name: e.g., siva-image. • Description: For backup or scaling purposes. • Click Create Image.

Step 5: Create a Launch Template

• Go to EC2 → Launch Templates → Create Launch Template. • Template Name: e.g., siva-template. • Description: "This is my launch template." • Choose your AMI: Select the image created earlier (siva-image). • Instance Type: t3.micro. • Key Pair: linuxx. • Network Settings: Select your VPC and subnets. • Click Create Template.

Step 6: Create a Classic Load Balancer (CLB)

• Go to EC2 → Load Balancers → Create Load Balancer. • Select Classic Load Balancer. • Name: CLB. • Select your VPC. • Choose Availability Zones: ap-south-1a, ap-south-1b, ap-south-1c. • Attach your Security Groups. • After creation, note the DNS name (e.g., CLB-xxxx.ap-south-1.elb.amazonaws.com).

Step 7: Create an Auto Scaling Group (ASG)

• Go to EC2 → Auto Scaling Groups → Create Auto Scaling Group. • Name: alg. • Select your Launch Template: siva-template. • Network: Select your VPC. • Availability Zones: ap-south-1a, ap-south-1b, ap-south-1c. • Attach to an existing Load Balancer → Choose your CLB. • Desired Capacity: 3. • Scaling Limits: Min = 3, Max = 5. • Monitoring: Enable (optional). • SNS Notifications: - Create topic: siva. - Create subscription → Protocol: Email → Enter endpoint email address.

Step 8: Create an Application Load Balancer (ALB)

• First, create a Target Group: - Name: t.g-ssa. - VPC: Select your existing VPC. - Health Check Path: /index.html. • Then, create an Application Load Balancer: - Name: alb-ssa. - VPC: Select your VPC. - Choose Availability Zones: ap-south-1a, ap-south-1b, ap-south-1c. - Attach Target Group created earlier. - Click Create Load Balancer. • Finally, attach the ALB to your Auto Scaling Group.

Best Practices & Tips

• Always use at least two Availability Zones for redundancy. • Enable CloudWatch monitoring to track instance health and scaling events. • Implement Lifecycle Hooks to manage instance startup/shutdown gracefully. • Use Target Tracking Scaling policies for dynamic scaling based on metrics (e.g., CPUUtilization). • Regularly update AMIs with the latest patches and security configurations.